MILITARY SPECIFICATION

TRUCK, FIREFIGHTING; PUMPER TANKER, 3,500 GALLON WATER CAPACITY DIESEL ENGINE DRIVE, 76,000 POUNDS GVW 6 BY 6

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one firefighting pumper tanker truck with a capacity of 3,500 gallons of water.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified (see 6.2), the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

GGG-A-926 - Ax (Single Bit, Double Bit, Pick Head and Mattock Head).
GGG-B-101 - Bar, Chisel, Pinch, Pry, Wrecking, Digging, Tamping, Digging and Tamping, and Crowbars.

MILITARY

MIL-G-3787 - Glass, Laminated, Flat; (Except Aircraft).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4210

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STANDARDS

FEDERAL

FED-STD-297 - Rustproofing of Commercial (Nontactical) Vehicles.

MILITARY

MIL-STD-281 - Automobiles, Trucks, Truck Tractors, Trailers and Trailer Dollies; Preservation and Packaging of.
MIL-STD-831 - Test Reports, Preparation of.
MIL-STD-1223 - Nontactical Wheeled Vehicles, Treatment, Painting, Rustproofing, Identification Marking and Data Plate Standards.

2.1.2 Other Government documents, drawings, and publications. The following other Government document form a part of this specification to the extent specified herein.

DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Vehicle Safety Standards and Regulations.
Federal Motor Carrier Safety Regulations.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as Department of Defense adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

Z26.1 - Safety Code for Safety Glazing Materials for Glazing Motor Vehicles on Land Highways.

(Application for copies should be addresssed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 1931 - Fire Department Ladders, Ground and Aerial.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

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SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The truck shall be a cab forward, six wheel, six wheel drive, midmounted diesel engine, aft of the cab and forward of the water tank. Truck shall have a gross vehicle weight (gvw) rating of not less than 76,000 pounds (lb). The truck shall be equipped with a 3,500-gallon water tank, and related piping and other equipment and accessories as specified herein.

3.2 First article. When specified (see 6.2), the contractor shall furnish a truck for first article inspection and approval (see 4.2.1 and 6.4).

3.3 Standard truck, components, and accessories. Except as specified herein, the truck, components, and accessories shall be the manufacturer's standard or optional items which meet or exceed the requirements of this specification. All items shall be as represented and rated in the manufacturer's sales information, including special or mounted equipment. Sales information shall be limited to specifications and technical material identical to that furnished to authorized company representatives for selection of truck models and components.

3.4 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

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3.5 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.6 Performance. The fully equipped truck, with 3,500 gallon water tank filled to capacity, a crew of three men in cab (weighing not less than 175 lb each or equivalent deadweight), and utilizing number two diesel fuel or JP-5 fuel, shall be capable of the following:

a. Ascending in both forward and reverse gears, a grade of not less than 18 percent while the transfer case is engaged in low range.
b. Accelerating on a level road from a standing start to a speed of not less than 40 miles per hour (mph) in not more than 45 seconds.
c. On a level road, maintain a road speed of not less than 45 mph.
d. Ascending a 10 percent grade at a maintained 20 mph.
e. Maintaining 1.5 mph without vibration or engine stalling.
f. The noise level of the truck shall not exceed the values as specified in 3.6.2.
g. The truck shall develop 70 percent of maximum drawbar pull for 1 hour in low gear.
h. The carbon monoxide concentrations shall not exceed 100 parts per million in the cab during truck operation.
i. The truck shall have mobility characteristics for low speed operation over open fields, broken ground, gravel, and general uneven cross-country terrain. The truck shall traverse, in one pass, at low speed, terrain consisting of unflooded fine grained, dry to moist soils having a minimum cone index (CI) rating in the critical layer of no greater than 45 as specified in 6.5.
j. Starting and operating in ambient temperatures ranging from -40 degrees Fahrenheit (°F) to +125°F. Preheating accessories, such as glow plugs, cold winterization package and ether starting aids, shall be used for low temperature engine starting.
k. The service brakes shall control and hold the fully loaded truck on a 20 percent grade. Under all conditions of loading, service brakes shall stop the truck within 35 feet from a speed of 20 mph on dry, hard, reasonably level surfaces free from loose material. The service brakes shall be capable of fully recovering within 3 minutes at the end of the road test (3 miles of average 10 percent

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grades). A brake fade test shall show a deceleration rate of 14 feet per second at 35 mph to 0 mph at a stopping distance within 85 feet.

1. The parking brake shall hold the fully loaded truck on a 20 percent grade incline.

3.6.1 Altitude conditions. The truck shall satisfactorily deliver its rated pumping capacity at any elevation from sea level to 2,000 feet at an ambient temperature as specified in 3.6.

3.6.2 Sound level. The interior sound level shall conform to DoT Federal Motor Carrier Safety Regulations, section 393.94.

3.7 Functional characteristics. The truck shall incorporate such functional characteristics that will result in a practicable combination of the requirements specified herein with ease of production, safety, and comfort of personnel, accessibility for maintenance, and minimum requirements for special maintenance tools. All mechanical designs shall incorporate good human factors, engineering principles, and shall emphasize reliability, simplicity, rapidity, and straightforwardness of operation by personnel wearing protective clothing and under great stress.

3.7.1 Covers and access plates. Covers and access plates shall be provided where necessary, to facilitate inspection, servicing, repair, and adjustment. Covers and access plates shall be secured with the minimum number of fasteners compatible with sealing required. Fasteners shall be quick-release type wherever possible.

3.7.2 Protective features. In addition to accessibility, the installation of drain cocks for all air and fuel reservoir shall be protected from accidental opening. The placement of all drains shall safeguard against drainage of contaminants on other components and accessories. Magnetized drain plugs shall be furnished for all gearboxes and oil reservoirs. Design of protective features shall:

a. Incorporate protection for fuel tank filler neck, door handles, brake cocks, flexible cables, pipe runs, and similar items to minimize damage resulting from use of these items for footsteps.
b. Assure that oil and fuel leakage or spillage does not contaminate other components, such as hose reel(s), batteries, personnel heaters, wiring, and junction boxes during operation or servicing of the truck.

3.8 Weights, dimensions, and clearances.

3.8.1 Gross weight. The gvw rating shall be not less than 76,000 lb.

3.8.2 Dimensions and clearance. The truck shall conform to the dimensions and clearance specified in table I.

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3.8.2.1 Ground clearance. Ground clearance, no body structure shall be below the centerline of the trucks axle with the trucks fully loaded gross weight.

TABLE I. Dimensions and clearance.

Overall length, inches (not more than)	420
Overall width, inches (not more than) 1/	100
Overall height, inches (not more than)	132

3.9 Chassis components.

3.9.1 Frame. The chassis frame shall be of manufacturer's heavy duty construction and shall consist of longitudinal side rails with crossmembers, gussets located to provide adequate strength for the imposed loads, and maximum accessibility of all driveline components. Crossmembers shall be of either the rectangular or cylindrical, tubular type with box type weldment located at the truck frame to equalize stresses in all welds and in the crossmember assembly. The frame shall be designed to enable the truck to withstand all tests specified herein without permanent deformation. Composite basic frame structure, involving ferrous and nonferrous structural material, shall incorporate design features to minimize the effects of corrosion.

3.9.2 Tow hooks. Two heavy-duty, forged steel tow hooks shall be mounted at the front and rear of the truck. The hooks shall be attached to the chassis frame members or to rigid members which are attached to the frame.

3.9.3 Engine. The truck engine shall be a Caterpillar Model 3406 turbo-charged diesel engine having the horsepower (hp), torque, and speed characteristics to satisfatorily meet all vehicular and pumping performance requirements specified herein. The engine shall be midmounted. The engine shall operate effectively on JP-5 fuel.

3.9.3.1 Air cleaner. The air intake for combustion shall be equipped with an oil bath or dry type air cleaner. The air cleaner, when tested in accordance with SAE J726, shall exhibit a cleaning efficiency of not less than 96 percent at a low engine air flow demand, with the efficiency increasing to not less than 97 percent in the 30 to 10 percent range of engine air flow demand. The cleaner shall exhibit no oil pullover characteristics at 100 percent engine air flow demand with the truck traversing cross-country terrain. The air cleaner shall be accessible and easily removed for cleaning.

3.9.3.2 Oil filter. The engine manufacturer's standard full-flow oil filter of the replaceable cartridge type shall be provided.

3.9.3.3 Governor. A governor shall be provided that will limit the engine to the maximum full load revolutions per minute (rpm) recommended by the engine manufacturer. The governor shall provide maximum effectiveness under all engine operating conditions and shall not restrict engine power output below governed full load rpm.

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3.9.4 Engine hour meter. An engine hour meter having a totalizing mechanism of not less than 9,999 hours shall be furnished for the chassis engine to register accurately the number of hours of operating time. The meter shall be of rugged construction to insure continuous trouble-free performance under severe operating conditions. The engine hour meter shall be mounted on the cab instrument panel or in the engine compartment in a readable location.

3.9.5 Cooling system. The cooling system with circulating pump shall be of sufficient capacity to prevent coolant temperature exceeding 225°F in a pressurized closed system, as measured at the head outlet when operating continuously under normal operating conditions in an ambient temperature of 125°F. The radiator core shall consist of materials and construction in accordance with the best commercial practice, and shall be mounted so that it will not be damaged by vibration or frame distortion. An expansion tank of sufficient capacity shall be provided for the expansion of the coolant in the engine water jacket, the radiator, heat exchanger, and the heating system provided. Provisions for air separation shall be included.

3.9.5.1 Heat exchanger. An auxiliary heat exchanger shall be provided which shall permit delivery of water from the fire pump to the heat exchanger for cooling of the engine liquid coolant without admixture. The heat exchanger shall be connected to the cab heater circuit. The heat exchanger shall be made of brass and shall be separate from the radiator. The discharge water from the heat exchanger shall be piped into the suction side of the pump. The size of the heat exchanger discharge line shall be large enough to effectively minimize back pressure on the heat exchanger. The flow of water from the fire pump to the heat exchanger shall be controlled by a valve, with the valve control extended to the equipment operators control panel.

3.9.6 Exhaust system. The exhaust system shall consist of an exhaust manifold, pipe(s), and engine turbocharger. Means shall be provided for self-draining of the exhaust system when truck is parked in a level position. Location of the exhaust system shall eliminate the potential hazard of heat, carbon deposits, and fumes which are likely to adversely affect other components or operating personnel. Exhaust system shall be constructed of suitable corrosion-resistant materials.

3.9.7 Fuel tank and lines. Fuel tank of not less than 50 gallons shall supply all fuel requirements. The tank and lines shall be amply protected against damage, excessive heat, and prevention of vapor lock. The tank shall be so mounted that it will not be damaged by frame distortion or collision with road obstructions, and shall not be affected by heat or flames that may surround the truck or from heat from the exhaust system. The fuel tank filler cap shall be the captive chained type. A compartment door shall provide easy access for insertion of filling nozzle. The fuel tank shall be equipped with a standpipe, 3/4-inch clearance from bottom of tank, for outlet fuel source to engine. Fuel tank shall have a drainage outlet located at the lowest part of

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bottom of the tank. Separate fuel pumps shall be provided for all components on the truck requiring diesel fuel. Fuel filters shall be included in the piping to each unit to prevent passage of foreign material.

3.9.8 Transmission and torque converter. A Allison Model HT740 transmission with converter clutch lockup and power take-off (PTO) feature shall be provided. A geared automatic or semiautomatic transmission, in conjunction with a torque converter, shall be provided. The transmission shall be as defined in SAE J645. The transmission shall provide for driver selection of individual gears or ranges of gears required for efficient operation throughout the speeds specified under the mobility requirement herein. The shift control shall incorporate a positive indicator for each gear or range selected. The transmission shall be so protected that the operator cannot inadvertently shift into reverse. The transmission assembly shall be of a commercial design and capacity that has been produced and supplied as part of a commercial or military vehicle, and shall have been field tested as a specific engine and transmission package. There shall be a minimum of four forward and one reverse speed ratios, properly spaced to provide the performance requirements specified.

3.9.8.1 Transmission capacity. The torque input capacity of the converter transmission assembly shall be in excess of the net torque output of the engine, and the maximum speed output of the engine shall not exceed the rated input speed of the transmission. The converter, with transmission design, shall acknowledge net flywheel hp.

3.9.8.2 Torque converter. The torque converter shall be heavy-duty having a multiplication factor of not less than two to one at stall. The torque converter shall be equipped with a heat exchanger. The torque converter transmission combination shall be provided with an oil filter.

3.9.8.3 Converter gage. A torque converter temperature gage, suitably mounted on the instrument panel, complete with a flashing warning light that will be clearly visible to the operator when the temperature of converter fluid exceeds the safe operating temperature, shall be provided. The signal light shall be suitably identified with a permanent metal plate. An additional audible device shall be provided which will actuate the horn when the converter fluid temperature exceeds manufacturer's recommended safe operating temperature.

3.9.9 Transfer case/differential. A Rockwell T-228-D single-speed transfer case/differential with inner axle differential compensation shall be provided. The differential shall be of the automatic or manual type. The manual type shall be remotely controlled from within the cab. All differentials shall be easily engaged.

3.9.10 Universal joints. Universal joints shall be of the latest design and shall be identical make and type. They shall function satisfactorily under continuous operation at angles that will prevail with fully equipped and loaded truck during all operational conditions.

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3.9.10.1 Power train. All components utilized in the power train shall have torque capacity in excess of the maximum torque imposed by any gearing and the prescribed service conditions.

3.9.11 Axles. The axles shall be of the full floating type and of such a design and construction that in the event of an axle shaft failure at any point, the wheel will not be released. The rear axle shall be a Rockwell SU170 tandem, with 58,000 lb gross axle weight rating (GAWR). The front axle assembly shall be a Rockwell FDS-1800 single reduction steering type, with 18,000 lb GAWR. The front axle design shall preclude objectionable steering reaction of loud popping noise and jerking during tight turns. In order to achieve self-aligning and anti-shimmying effects, the steering axle shall be installed with positive caster and proper toe-in. The axles shall be designed for wide track, single tire mounting.

3.9.12 Suspension. The springs shall be heavy-duty, of adequate strength to sustain the maximum rated gross weight without evidence of overload or permanent set. Shock absorbers shall be furnished for the front and have a minimum capacity as required for the imposed load. Auxiliary springs shall be provided, if required.

3.9.13 Power steering. Power assisted steering shall be provided and shall be such that a torque, not exceeding 15 lb applied at the steering wheel, will produce maximum efficiency of the power assisted steering at low engine speeds.

3.9.14 Wheels, rims, and tires. The truck shall be equipped with single front and rear wheels. Rim size and tire ratings shall conform to TRA recommendations. Tires shall be of the wide base, and of rated capacity at least equal to the load imposed on each tire, measured at each wheel, at the ground, with the truck loaded to rated gvw and operating at maximum road speeds. A spare wheel with mounted tire shall be furnished, but shall not be mounted on the truck.

3.9.15 Braking system. The truck shall be equipped with all-wheel, full air application brakes with quick air brake buildup system. The braking system shall be capable of safely controlling the fully equipped and loaded truck under all operating conditions. The system shall be pedal operated from the driver's position and conveniently located so that the driver, wearing rubber boots, can safely activate the brake pedal without interference from other control pedals or cab obstructions. Ease of accessibility for servicing and adjustment of brakes is essential. The design shall include adjustment features without removal of the wheels, and shall include shields to safeguard against accumulation of mud and water in the brake drum. The braking system, complete with all necessary components, shall include:

a. A 12 cubic feet per minute minimum rated compressor with an oil bath or dry air cleaner on the intake side.
b. Air storage reservoir(s) with not less than 2,400 cubic inch capacity, each tank equipped with drain and with safety and check valves between compressor and last reservoir tank.

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c. Foot control, relay, and safety valves.
d. Air pressure gage visible to the driver.
e. Low pressure warning indicator, visible and audible to the driver.
f. An Anchorlok air dryer of at least four stage process to effectively remove water, oil carbon, and sludge from the air brake system shall be provided.
g. Automatic slack adjusters.

3.9.15.1 Air brake parking system. The vehicle shall be furnished with full air release-spring applied parking brake system. The system control to release and apply the parking brake shall be accessable by the operator seated in the driver's seat. A manual or mechanical means to release the brakes shall be provided when compressed air is not available to release the brakes. The parking system shall conform to DoT Federal Motor Carrier Safety Regulations, section 393.41.

3.9.16 Electrical system. The electrical system shall be 12 volts (V) nominal direct current (dc) potential. Wiring shall be thoroughly secured in place, shall be insulated, splash proofed, and protected against exposure to ground fires. The electrical system shall include an alternator-rectifier system, batteries, switches, circuit breakers, lighting, a red and blue light bar, an electronic siren-speaker system, and horn.

3.9.16.1 Alternator. An alternator-rectifier system shall be furnished complete with voltage regulator. The alternator shall be rated for not less than 145 ampere (A).

3.9.16.2 Battery(s). Battery(s) furnished shall be 12V potential, and shall be rated in terms of cold cranking A capacity at 0°F in accordance with SAE J537. The battery system shall have a total cold cranking capacity of not less than 1,100 A, with a reserve capacity of 445 minutes. The battery(s) shall be easily accessible for servicing and removal.

3.9.16.3 Lighting. The truck shall be equipped with clearance lights, parking lights, stoplights and taillights, and directional signals. All lights shall be of sufficient size and shall be in accordance with DoT Federal Motor Carrier Safety Regulations. In addition, the lighting shall conform to the following:

a. Two sealed beam headlamps. Headlamps shall be mounted as low as practicable, beams shall be adjustable and shall be adjusted so that center of upper beam will strike the ground 300 feet ahead of fully loaded truck on level ground.
b. Directional signal lights shall be of the self-canceling type with visual and audible indicator. The controller shall be mounted on the steering column and shall be wired through the ignition system. A switch for 4-way emergency flashing shall be provided.
c. Cab dome lights with independent switches shall be provided.
d. All gages and valve controls shall be properly lighted and readily visible from the operator's position.

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e. A red and blue light bar shall be installed on the top front area of the truck so as to provide maximum visibility over 360°. The control switch shall be installed on the cab instrument panel and be independent of the dash switch.
f. All lenses on front mounted headlights shall be glass 1/.
g. An engine compartment light(s), with independent switches shall be provided.
h. One six inch diameter, double face, red colored, flashing light, shall be installed on each top rear corner of the water tank. The control switch shall be installed on the cab instrument panel and be independent of the dash switch.

NOTE: 1/ The lenses on the front mounted lights (turn signals, reflectors, and marker lights) may be other than glass providing they have the highest temperature melting characteristics available.

3.9.16.4 Siren and speaker system. A combination electronic siren and external speaker system shall be mounted on the top front right corner area of the truck. The unit shall be Federal PA15A, and shall be controlled within the cab and within reach of the driver.

3.9.16.5 Horn. An electrical horn actuated by a button in the hub of steering wheel shall be provided.

3.9.16.6 Wiring. All wiring shall be encased in conduit, loom, or wiring harness, and shall be color coded or number identified. The harness or individual wiring shall be protected by the use of grommets where they pass through metal bulkheads.

3.9.16.7 Backup alarm. The vehicle shall be provided with an audible electronic, pulsating, signaling device to caution personnel when the vehicle is in a reverse gear operation.

3.9.17 Instruments and controls.

3.9.17.1 Driver's instrument panel. Instruments and controls shall be grouped by function insofar as practicable and shall be located so that they can be seen from the driver's position. All instruments and contols shall be suitably identified by metal plates with letters at least 1/4-inch high in contrasting colors. All controls shall be capable of effective operation by the 5th through 95th percentile of personnel wearing crash clothing and rubber boots. The main instrument panel shall contain as a minimum the following:

a. Speedometer and recording odometer.
b. Tachometer for the engine.
c. Ammeter or voltmeter.
d. Fuel gage.
e. Coolant temperature indicator.
f. Oil pressure gage for engine and transmission.

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g. Oil temperature gage for transmission.
h. Ignition and starting switch.
i. Master switch for accessories.
j. Light switches except floodlights, compartments, and trouble lights.
k. Headlight beam indicator.
l. Windshield wiper and washer controls.
m. Low air pressure signal.
n. Air pressure gage, green and red sector.
o. Pump engage control system.
p. All instruments shall be lighted indirectly and controlled by a separate switch so that the instruments can be lighted without turning other lights on.
q. All gages and instruments shall be marked with red and green lines or sections as appropriate to indicate normal operating conditions or "do not exceed values."

3.9.17.2 Equipment operator's control panel. An equipment operator's control panel shall be provided and conveniently located outside the cab, at the street side (driver side) for housing all control instruments, gages, and instruction plates applicable to firefighting, water system performance as controlled by the operator. Commercially available gages which incorporate more than one function are acceptable. This control panel shall contain as a minimum:

a. Two inch weather proof oil pressure gage, to monitor engine oil pressure.
b. Two inch weather proof water temperature gage to monitor engine water temperature.
c. Water tank level indicator lights.
d. Engine throttle control.
e. Engine tachometer.
f. Gages, silicone filled type, 3-1/2 inch, one for each discharge connection and cross lay hose bed.
g. Pump engage indicator light.
h. Pump panel light.
i. Valve controls for all discharge and suction connections.
j. Drain valve control.
k. Control valves for cross lay hoses.
l. Pump primer control.
m. Pump pressure relief valve control.
n. Test plugs for testing vacuum and pressures.

3.9.18 Cab.

3.9.18.1 Cab construction. The cab shall be an all aluminum structure of welded construction, enclosed forward single compartment with ample room for a three-man crew. Interior of body sides, roof, and doors shall be treated with an insulating material and shall be lined with washable, high strength, pressed wood fiber, marlite, plastic, or metal panels. The access steps to cab

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interior and exterior shall be so designed as to provide maximum safety for personnel wearing crash clothing when entering or leaving the cab. The cab shall be rainproof and dripproof, and shall provide for maximum visibility. The cab shall be tested to a water spray simulating a rainfall of 3 inches per hour for period of 1 hour to determine leakage.

3.9.18.1.1 Cab doors. The cab shall have one door on each side; all cab doors shall be designed so that fire fighting personnel wearing foul weather hand gloves can open and close the doors with minimum amount of effort. Automatic locking of the doors in the fully opened and closed position with manual release shall be provided. All doors shall have design features which will prevent the doors from becoming damaged under any condition of vehicle operation required herein. The outside door handles shall be not more than 6 feet above ground level and shall not protrude appreciably beyond the body skin.

3.9.18.1.2 Cab seating. A seat shall be provided for the driver and shall be a Bostrom heavy duty Viking T-Bar. Bench type seating spaces for two crewmen shall be provided in the cab. Bench type seats shall be padded seats of suitable width and depth to provide ample seating width for crewmen in turnout coats. All seats shall be upholstered over polyurethane foam cushion with first grade Naugahyde suitable for fire service. Seat belts shall be furnished and installed for the driver and for each seated position provided. Seat belt length shall be made to accommodate crewmen wearing turnout coats. Storage space shall be provided under crewmen seats.

3.9.18.2 Windows. The cab windshield shall be laminated, safety glass conforming to MIL-G-3787, class II. The width of the windshield shall be the maximum consistent with the truck width. Windows of tempered, safety glass per ANSI Z26.1 shall be provided in the cab and doors. All windows shall be capable of sliding open or rolling down.

3.9.18.2.1 Visibility. The design and arrangement of the cab and operating components contained therein shall be such as to afford optimum visibility for control of the truck and performance of firefighting operations. The operator shall be able to see the ground at a point not greater than 20 feet from the truck, through an included angle of at least 105° on each side of the centerline of the windshield (210° total angle). The restriction of the horizontal angle of vision by obstructions, such as window frames, corner, and door posts shall not exceed 7° each. Forward vision through the windshield between the forward corner posts of the cab shall be unobstructed, except the center post between the operator is permissible if the center post width is restricted to that width (in all cases under 2-1/4 inches) which imposes no blind area at a distance of 40 feet from the truck when viewed by either the driver or crew member sitting in a fixed, normal operation position. Upward visibility at least 15° above the horizontal is required at eye level.

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3.9.18.3 Windshield washers and wipers. The windshield wipers shall be heavy-duty dual wipers of the electrical type with capacity and coverage to readily cleanse the windshield of rain, snow, and ice, when used in conjunction with defrosting system or windshield washer mechanism. Wiper arms and blades shall be of sufficient length to clean at least 70 percent of the windshield area. Wiper assemblies shall be synchronous and shall include features to permit speed control and return of the wiper blades to a retracted position out of the line of vision when not in use. The washer reservoir shall have a capacity of at least 1 gallon and the container must be nonbreakable. All controls shall be in reach of the driver from his normal driving position.

3.9.18.4 Heating system. A coolant type heater shall be provided in the heating system. Heater shall operate satisfactorily on diesel fuel supplied from the truck engine fuel tank. The liquid distribution system shall be adequate to protect personnel, water, and equipment during exposures to ambient temperatures of -40°F. Heat output for the cab shall be sufficient to provide and maintain a temperature of +40°F at the driver and crew's feet position at an ambient temperature of -40°F after 1 hour of heater operation. Adequate heat to maintain a temperature of +40° shall also be supplied to the remainder of the truck to protect those water tank, lines, and equipment components subject to freezing during standby operation at -40°F. Fluid lines shall not be drained during these operations. Cab heating system shall meet the requirements of DoT Federal Motor Carrier Safety Regulations, section 393.77, as applicable.

3.9.18.4.1 Power plant heater. An engine coolant heater shall be provided. The heater shall operate on 110V, alternating current (ac), 60 Hertz (Hz), and shall be wired through a junction block to a single three pronged (male) weatherproof slave receptacle for receiving external power and grounding vehicle. The receptacle shall be located on the pump operator control panel. A three-wire connecting cable, 25-feet long, and of adequate capacity to supply power to the heater, shall be furnished. The connecting cable shall include a matching female connector at the vehicle end, and a standard three-pronged (two power plus one ground) male connector at the other end. Electrical apparatus shall conform to DoT Federal Motor Carrier Safety Regulations, section 393.77(c), (7). The electrical insulation of the connecting cable shall withstand normal operating stresses in low ambient temperatures (down to -65°F) without cracking or loss of dielectric capacity. All heater lead wires shall be installed without interfering with vehicle component operation and without loose excess wire. Provisions for stowage of the cable shall be provided in the vehicle cab. The heater furnished shall be as follows:

a. A coolant heater shall be installed in the engine block or lower coolant hose. A coolant circulating pump driven by a 110V, ac, 60 Hz motor shall be provided. The coolant heater shall be a 2,250 watt, 19.5A, 115V, 60 Hz, ac, heater, Kim "hot-start", Model BC-110.

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b. Immerson type engine oil heater, 300-watt minimum rating, with 170°F to 195°F thermostat, shall be installed in oil pan through any convenient opening.
c. Battery heater shall have a capacity to maintain the battery electrolyte at a temperature of not less than 10°F during vehicle exposure to ambient air temperatures as low as -60°F, and shall embody a thermostat to limit the temperature of the electrolyte to not more than 80°F.
d. A fuel warmer or preheater shall be provided on diesel engine driven vehicles to prevent clogging of fuel filters due to wax crystallization in the fuel. The fuel warmer shall use engine coolant to transfer sufficient heat to the diesel fuel to heat it from an inlet temperature of -40°F to an outlet temperature of +9°F, with a flow rate not less than the maximum fuel demand of the engine fuel system. A coolant shutoff valve shall be provided for the coolant inlet side of the fuel warmer unit.

3.9.18.4.2 Fuel fired engine preheater. A fuel fired engine heater shall be furnished to preheat the engine. The heater shall include a timer, a thermostat, and a 12 V dc electric water circulating pump and shall be connected directly across the engine coolant inlet and outlet. The heater shall be capable of starting and operating at -60°F and shall heat the engine to +40°F from -60°F in not more than 1 hour. The system shall be equipped with a light, visible to the driver, to indicate that the preheater is operating.

3.9.18.4.3 Defrosting, anti-icing, and ventilating. A sufficient defrosting, anti-icing, and ventilating system shall be provided in the cab. This system shall have the capacity to effectively prevent formation of frost in the interior surface of all transparent areas of the cab and to prevent the exterior formation of ice on the windshield of the vehicle. The system shall also maintain clear vision over at least 75 percent of the area of all windows when the cab interior is at a mean temperature of 40°F and the ambient temperature is -40°F. Fresh air intakes, louvered from the outside, shall be provided as a means of diverting the output air to a point outside the cab in order for infinite heat input control and prevent overheating with moderate outside temperature.

3.9.18.4.4 Controls. The heat supplied to body area, protecting water, lines, and equipment components, shall be thermostatically controlled from 40°F to 75°F. Variable speed controls shall be provided on air blowers and at least high and low output settings shall be furnished on cab heating system controls. Adequate operating instructions shall be provided at heater system control point. Cab heater and defroster system control shall be mounted on the instrument panel.

3.9.18.5 Electrical heated rearview mirrors. An external combination (flat and convex) mirror, enclosed in a common housing and each with at least four supporting arms or heavy-duty, spring loaded, ratchet type mounting

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brackets, shall be mounted on each side of the cab. Manufacturer's electrical heated type mirrors assembly shall be provided and, each combination mirror shall have at least 50 square inches of flat reflective area and an adjustable convex surface having at least 20 square inches of reflective area. Each mirror shall be viewable from the operator's position.

3.10 Body. The body shall consist of a section which shall enclose the engine/transmission assemblies. The body shall be made of aluminum alloy having not less than 42,000 pound-force per square inch (psi) ultimate and 35,000 psi yield strength and a thickness of not less than 0.050 inch. The body section shall be insulated and heated to assure that the temperature will not fall below 40°F with an ambient still air temperature of -25°F. All body insulation shall be enclosed. Fixed steps shall be provided and constructed of nonskid material, steps shall be conveniently located on each side of the body. Engine service entrance door(s) or panel(s) shall be provided for ease of maintenance of the engine and transmission.

3.10.1 Compartments. Suitable compartments shall be provided for the convenient storage and removal of equipment and tools. Compartments for the storage of items which need exposure protection shall be weathertight, insulated, and heated. Doors shall be reinforced and of slam-shut design. Operating handles shall be of a chromium-plated steel, operable with heavy-gloved hands. Single hinge on doors shall have spring operated props for one-hand operation. Drip rails shall be installed above all compartment doors. All compartments shall be provided with adequate lighting and the lighting shall be controlled by door switches. Provisions shall be incorporated to permit flushing and cleaning inside of compartments. The compartment to be provided, shall be as follows:

a. Compartment for storage of one double crosslay hose bed, with 250 feet by 1-1/2-inch double jacket fire hose, shall be located above the control panel area.
b. Two storage compartments approximately 27 inches wide by 27 inches high by 20 inches deep and shall be located on the street side of the truck, ahead of the intermediate wheel area.
c. Two storage compartments approximately 27 inches wide by 27 inches high by 20 inches deep and shall be located on the curb side of the truck, ahead of the intermediate wheel area.

3.10.1.1 Stowage equipment. An enclosed compartment shall be provided for stowage of accessory equipment.

3.10.1.2 Transverse hose beds. Two transverse hose beds shall be provided. Each bed shall have the capacity of stowing not less than 250 feet of 1-1/2-inch double jacketed, polyester rubber lined fire hose. Each hose bed shall have hose payout openings on the street side of the truck. The length of each transverse hose bed shall be constructed of aluminum. Each bed shall have a sub-floor made from slatted aluminum. Each hose bed shall be provided with a 1-1/2-inch swivel type discharge connection to permit hose to

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be removed from street side side of the truck. There shall be adequate clearance for making connections to the swivel discharge outlets. Discharge gate controls for each hose bed shall be located at the equipment operator's control panel and shall be suitably identified. All hose payout openings shall be fitted with 1 1/4-inch chromium plated hose rollers on the bottom and sides of the openings. Adequate drains shall be provided to each transverse hose plumbing.

3.11 Firefighting system and equipment.

3.11.1 General. The firefighting system shall consist of a water tank, water dispensing outlets, water pump with appropriate piping, and system controls to meet the performance requirements specified herein.

3.11.1.1 Firefighting system performance. The vehicles firefighting system, when in pumping mode, shall discharge water from storage tank at a rate of not more than 500 gallons per minute (gpm) through the discharge system.

3.11.2. Fire pump. The unit shall incorporate a Hale QFLA two-stage centrifugal fire pump for water application, and be PTO driven with pump and roll capabilities. The suction inlet(s) of the pump shall be lower than the bottom of the water tank to assure gravity flow to the pump under all conditions specified herein. The pump shall be removable as an assembly without disturbing other major components. The pump shaft shall be fabricated of corrosion-resistant steel and the impeller shall be bronze. The pump casing shall be cast iron or aluminum.

3.11.2.1 Pump system controls. The control system for the pump, shall be manually operated. All components shall be capable of being operated manually in case of loss of pressure. Manual controls to valves must be easily accessible with adequate performance clearances, free from encumbrance, and so located to permit personnel to reach and manually operate control levers during emergency situations. Manual controls must be identified at the valves with "on-off" operating procedures clearly marked. All manual controls shall be located at the pump panel area and shall be accessible for one man operation.

3.11.2.2 Pump inlet piping. Inlet piping for the water pump shall provide for continuous flow of liquid to the pump without cavitation at all flow rates up to rated system capacity. Tank shutoff valve(s) shall be provided in the inlet piping of the water pump. The water tank shutoff valve(s) shall be controlled at the left side of the truck.

3.11.2.3 Water pump automatic pressure control device. The pressure control device shall incorporate safety features as per NFPA No. 1901. The pressure control device shall ensure that proper system pressure is maintained for operation. A relief valve control wheel and indicator light shall be mounted on the operator's pump panel.

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3.11.2.4 Pump discharge connections. Two 2 1/2-inch national hose (NH) male discharge connections shall be located on the curb and street side of the truck and shall terminate near the operator's control panel as specified in 3.9.17.2. The connection shall incorporate a quarter turn valve with nonsticking ball and self-adjusting seats. The valve shall be self-locking in any position. The connections shall be provided with a chained-on brass or chromium-plated brass rocker lug cap. The connections shall be no more than 48 inches above the ground. Piping from the fire pump to the connections shall be sized corresponding to the size of the connections without the use of reducers.

3.11.2.5 Pump suction connections. The water pump inlet piping shall include a 4 1/2 inch NH suction valve(s). The valves shall be quarter turn, nonsticking ball with self-adjusting double seat to hold pressure or vacuum in either direction. The suction valve(s) will have female swivels with removable strainers, self-locking handles, and chained-on brass or chromium-plated brass rocker lug caps. One pump suction connection shall be located in the side operator's control panel and the other shall be located at curbside opposite control panel. The connection shall be used for pumping from a resupply vehicle or from a hydrant. Piping to the fire pump from the connection shall be sized corresponding to the size of the connection without the use of reducer. 4-1/2 national hose steamer inlets for use with suction hose shall be installed on each suction side of the pump.

3.12 Water tank. A water tank of not less than 3,500 gallon capacity shall be provided. The water tank shall be fabricated of aluminum alloy. The water tank shall be provided with a sacrificial anode system to neutralize the water condition and to protect the tank against corrosion. The water tank shall be of the horizontal type, with flat top and flat bottom, with sump, and mounted on the truck frame. A longitudinal and transverse surge partition shall be provided in the tank. A 2-1/2-inch connection shall be provided at the pump panel and piped directly to the booster tank for use as a direct supply connection, it shall enter, the tank or be piped inside the tank so as to not cause cavitation at the tank to pump discharge. The tank drain shall be located at the low point of the tank to provide complete drainage and the drain control valve shall be operated/controlled from the pump panel. The water tank drain shall discharge directly to the ground at a point not above the bottom of the chassis frame. The tank shall have a nonsplash combination overflow and vent pipe of not less than 4-inch National Pipe Size, directed to the ground at the rear of the rear axle, and extending below the bottom of the truck frame. The tank shall have a warranty for 5 years. A permanent installed rear mounted access ladder shall be provided for the operator, and be conveniently centered with the tank filler opening.

3.12.1 Tank filler opening. The tank shall have a filler opening not less than 6 inches inside diameter and shall be provided with a tight, gasketed, hinged cap having a spring-type holddown catch. The tank fill opening shall be provided with a removable strainer. The tank filler opening shall be conveniently located at the top, rear of the water tank.

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3.12.2 Tank piping. The piping from the tank to the fire pump, unless made of corrosion-resistant metal, shall be galvanized. The pipeline between the tank and the pump shall be adequate in size to permit flow rate of 750 gpm and shall incorporate a flexible, reinforced, noncollapsible connection(s) to take up vibration. The tank piping shall provide for discharging the water in the tank through the fire pump discharge manifold. A quick-opening shutoff valve shall be provided with control extended to the operating side of the truck and a drain valve located at the lowest point with control extended to the operating side of the truck.

3.12.3 Tank mounting. Tank shall be supported on sills to cushion shock and transfer the load to the truck frame. The tank shall be so mounted and attached to the chassis that it can be removed without removing the truck body.

3.12.4 Tank level gage. Tank level gage shall be provided for the water tank. The gage shall be mounted on the operator's pump panel. The gage shall be a M.C. Products five light type. The gage shall allow a determination of quantity of water in the tank with an accuracy of $\pm$5 percent of the tank capacity at any level from empty to full.

3.13 Winterization of lines. All piping and components subject to freeze damage when the truck is stored in cold weather shall be provided with adequate heat wrapped tape.

3.14 Accessory mounting brackets and holders. When specified (see 6.2), brackets or quick-release holders shall be furnished and installed for mounting the following accessories (accessories are to be furnished):

a. One 35-foot length extension ladder, conforming to NFPA No. 1931.
b. One crowbar, conforming to GGG-B-101, type II, class 1, size 4.
c. One fire axe, conforming to GGG-A-926, type II.
d. One fire axe, conforming to GGG-A-926, type I, class I, design A.
e. 500 feet of 1-1/2 inch, double jacketed, polyester rubber lined fire hose.
f. Two, 1-1/2 inch turbo jet nozzles.

3.15 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat and type of paint materials shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag, and spatter, loose mill scale, water, dirt, corrosion product, or any oher contamination substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current

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materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.16 Lubrication. Means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high-pressure lubricating equipment, 1,000 psi or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location. The unit shall be lubricated prior to delivery with type of lubricant specified in the operator's manual and grade of lubricant recommended for ambient temperature at the delivery point. The unit shall be conspicuously tagged to identify the lubricants and their temperature range.

3.17 Servicing and adjusting. Prior to acceptance of the truck by the Government, the contractor shall service and adjust the truck for immediate operational use as required in the operator's manual. The servicing and adjusting shall include at least the following:

1. Inflation of all tires.
2. Adjustment of brakes.
3. Proper functioning of all lighting and electrical systems.
4. Wheel alinement.
5. Adjustment of engine to include tune-up.
6. Complete lubrication with grades of lubricants recommended for ambient temperature at the delivery point.
7. Cooling system filled to capacity with a clean solution of equal parts by volume of water and antifreeze (ethylene glycol).

The truck shall be conspicuously tagged to identify the lubricants and their temperature range.

3.18 Electromagnetic interference characteristics. The truck chassis shall conform to the electromagnetic characteristic requirements and test limits of SAE J551.

3.19.1 Rustproofing. The vehicle shall be rustproofed in accordance with FED-STD-297.

3.20 Marking and data plates.

3.20.1 Identification plate. The contracting officer will furnish to the Government inspector the required identification plates. The contractor will be required to stamp the necessary data in the blank spaces thereon and securely affix said plates in a conspicuous place on each unit, assembly or subassembly, and parts as directed by the Government inspector. Nonferrous screws, rivets, or bolts of not less than 1/8-inch in diameter shall be used

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to affix the plates. Nomenclature shall be "TRUCK, FIREFIGHTING; PUMPER TANKER, 3,500-GALLON WATER CAPACITY." Identification plates shall be in accordance with MIL-STD-1223.

3.20.2 Identification marking. Identification shall be permanently and legibly marked directly on the truck or on a corrosion-resisting metal plate securely attached to the truck at the source of manufacturer. Identification shall include the manufacturer's model and serial number, name and trademark to be readily identifiable to the manufacturer.

3.20.3 Instruction plates. The truck shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous screws or bolts of not less than 1/8-inch diameter.

3.21 Lifting and tiedown attachments. The truck shall be equipped with lifting and tiedown attachments. Lifting and tiedown attachments shall conform to type II or type III of MIL-STD-209. A nonferrous transportation plate shall be provided and mechanically attached to the truck. Transportation plates shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the item furnished showing the center of gravity shall be provided on the transportation plate. Tiedown attachments may be identified by stenciling or other suitable marking. Tiedown marking shall clearly indicate that the attachments are intended for the tie down of the truck on the carrier when shipped.

3.22 Workmanship.

3.22.1 Material fabrication. The material used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.22.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.22.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

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3.22.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on the truck when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.3 and the tests of 4.4 through 4.8.5. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.10, and the packaging inspection of 4.11.

4.3 Examination. Each truck shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual

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examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Tests.

4.4.1 Preparation for test. The first article truck submitted for test shall contain all specified components and shall be completely assembled and properly prepared for operation. The truck, including all components, shall be correctly adjusted, lubricated, and serviced, and all equipment which has been ordered shall be mounted for immediate operation.

4.4.2 Data. During the tests specified herein, at least the following data, unless not applicable, shall be recorded at intervals not greater than 15 minutes or as otherwise indicated. Additional data may be required and recording intervals decreased for test of short duration or as otherwise dictated by test objectives.

a. Date.
b. Time started.
c. Time finished.
d. Ambient temperature.
e. Atmospheric pressure.
f. Level in the water tank.
g. Water tank temperature.
h. Water pump inlet pressure (in Hg).
i. Water pump discharge pressure.
j. Engine speed (rpm).
k. Engine oil temperature.
l. Engine water temperature.
m. Torque converter oil temperature.

4.4.3 Test observations. Throughout all tests specified herein, the truck shall be closely observed for the following conditions which shall be considered cause for rejection.

a. Failure to conform to performance requirements specified herein.
b. Spillage or leakage of any liquids including coolants, and lubricants under any operating condition.
c. Structural failure of any part including permanent distortion or evidence of impending failure.
d. Evidence of excessive wear.
e. Interference between truck components or between the truck and the ground and all required obstacles with the exception of normal contact by the tires.
f. Misalignment of parts and components.
g. Instability in water system pressure of flow.
h. Evidence of undesirable roadability features including instability in handling during turning, braking, and when traversing rough terrain.

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i. Conditions which present a safety hazard to personnel during operation, service, or maintenance.
j. Overheating of the truck engine or any components.

4.5 First article test report. Upon completion of the first article tests, a test report shall be prepared in accordance with MIL-STD-831.

4.5.1 Reliability and maintainability information. The following information shall be included as an appendix to the test report:

a. All failures, servicing, adjustments, maintenance, and irregular functioning shall be identified by accumulated operating time, cycles, miles, or position in the test procedure as appropriate. Test conditions at the time of the events identified shall be recorded.
b. A summary of the engineering analysis and of any test conducted to determine assignable causes for any failure or irregular functioning.
c. A summary of the engineering analysis leading to any corrections made to design, construction, quality control, or other procedures or leading to any corrections to be made or proposed to be made to production items. The summary shall also include an analysis of the predicted effectiveness of such corrections.
d. Clock time and man-hours required for each maintenance and servicing action taken during the tests.

4.6 First article firefighting system tests.

4.6.1 Features and components. All features and components shall be inspected for proper installation, adjustment, and serviceability. The truck shall be driven for a distance of at least 5 miles over paved highway and at an average speed of 40 mph and shall exhibit no malfunctioning, leaks, objectional rough operation, or irregular movement.

4.6.2 Water tank.

4.6.2.1 Water tank filling. The empty water tank shall be filled from a 2-1/2-inch hose placed in the filler opening at an average flow rate of at least 250 gpm. The filler opening shall be checked for proper size strainer. The quantity of water required to fill the tank shall be measured. At the end of the filling cycle the tank overflow device shall be checked.

4.6.2.2 Water tank side filling inlet (hydrant). It shall be demonstrated that the tank can be filled from pump suction connections located in driver's side compartment specified in 3.11.2.6.

4.6.3 Water discharge.. The truck shall be operated to demonstrate that the water system will discharge water from the discharge outlets at a flow rate of at least 500/750 gpm; the two 2-1/2-inch discharge outlets shall have at least a flow rate of 250 gpm each and the two 1-1/2-inch tranverse hose

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discharge at least 95 gpm. The tests shall be conducted for each nozzle individually in a straight stream, widest disperse pattern. The water system shall be observed to determine that there is no instability in pressure or flow rate under any operating condition. Each flow condition shall be maintained for a minimum of 3 minutes.

4.6.4 Water tank exhaustion. The capability to discharge 97 percent of the water tank capacity by normal means when the truck is on level ground and 80 percent when the truck is headed in any direction on a 20 percent slope shall be demonstrated.

4.6.5 Transverse hose bed tests. The transverse hose bed shall be checked for proper stowage. Water discharge rates, and pattern characteristics shall be met as specified herein. During operations, the controls shall be checked for adequacy and ease of operation. The transverse hose bed shall be run fully out under simulated emergency conditions in order to determine adequacy of design and shall be stowed after use.

4.6.6 Auxiliary equipment. All portable extinguishers, ladders, and tools shall be removed from their mounting brackets in order to determine their ready availability.

4.6.7 Water drain system. The trucks water tank drain system shall be demonstrated for proper operation.

4.7 First article vehicular performance tests.

4.7.1 Weight determination. The gross and curb weights of the truck shall be determined when loaded as specified in 3.6.

4.7.2 Determination of load distribution. The loads on each wheel and each axle shall be determined at curb and gross weight for compliance with 3.8.2.

4.7.3 Determination of turning clearance and turning characteristics. A marker shall be so placed that it will describe the largest circle in a turn at maximum cramping angle. The truck shall then be moved with steering cramped at maximum angle to the left (marker located on the right side) until a circle has been inscribed on the ground. The diameter of this inscribed circle represents the turning clearance to the left. This procedure shall be repeated with the marker on the left side of the truck to obtain the turning clearance to the right. The truck with the water tank 75 percent full shall then be subjected to sharp turns at various speeds in both directions and the reaction of the truck observed.

4.7.4 Determination of limits of vision. The limits of vision of the driver crew member shall be determined by direct visions as specified in 3.9.18.2.1, seated in the truck, in their positions and using a large pad marked with concentric circle to indicate distance to right or left of normal. Blind spots within these fields of vision shall be noted.

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4.7.5 Vehicle performance tests. The total gross weight of the complete fire truck shall not exceed the manufacturer's gvw rating when fully equipped as specified herein, including all specified firefighting equipment serviced with fuel, lubricants; full of water capacity and a crew in the cab of three men weighing not less than 175 lb each or equivalent deadweight. The weight distribution shall be as specified herein. The truck, when tested as described below, shall comply with the performance requirement specified in 3.6. The truck shall be tested over prepared and selected test courses to determine the vehicular performance of the test item.

4.7.5.1 Maximum speed. The truck, with gross load, shall be operated on dry, level concrete roadway to determine maximum speed. A total of at least 10 test runs (five in opposite directions) shall be conducted. Maximum speed will be the average value obtained.

4.7.5.2 Acceleration. The time-speed relation of the truck with gross load shall be determined while accelerating from a standstill to its maximum speed on level concrete highway. A total of at least 10 test runs (five in opposite directions) shall be conducted. Acceleration will be the average value obtained.

4.7.5.3 Grade speeds. The truck, with gross load, shall be driven up a grade of not less than 18 percent, and the maximum speed shall be determined. Engine oil and fuel pressure shall be observed and engine starting and idling ability shall be determined. The truck is to ascend a 10 percent grade at a maintained speed of 20 mph.

4.7.5.4 Side slope operation. The 10 percent side slope operational characteristics (both right and left) of the truck with gross load shall be determined upon courses prepared for testing 10 percent side slope operation. Firefighting water discharge characteristics shall be determined during these tests. Any excessive leakage from water tank shall be noted. The vehicle shall be parked on a 20 percent grade and the engine shall be stopped, started, and idled for 2 minutes.

4.7.5.5 Drawbar pull and cooling. In an ambient air temperature not lower than 60°F, the truck with gross load shall be operated on dry pavement under road-load conditions until normal operating temperatures of the power train components have been reached. Utilizing a chassis dynamometer, the truck shall then be operated for 1 hour in low gear at 70 percent of maximum drawbar pull or until one of the component temperatures exceeds these critical values:

a. Engine coolant--95°F above ambient extrapolated to 125°F.
b. Engine oil--150°F above ambient extrapolated to 125°F.
c. Gear box oil--175°F above ambient extrapolated to 125°F.

Temperatures obtained shall be extrapolated on a one-for-one basis for 125°F.

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4.7.5.6 Braking ability. The truck, with gross load, shall be brought to a stop with the service brakes when headed both up and down a dry, hard surfaced, 20 percent grade. After the truck has been stopped when headed both up and down on a 20 percent grade, the parking brake shall be applied and the service brakes released. The truck shall remain stationary. The truck shall be driven at a speed of 20 mph on level concrete, and the minimum distance required to bring the truck to a stop shall be measured. This shall be accomplished by appropriate methods which will eliminate the reaction time of the driver. The capability of the service brakes to fully recover within 3 minutes at the end of a run, down a 3-mile (average 10 percent) grade with transmission in high gear and road speed controlled by repeated application of brakes not to exceed 30 mph, shall be demonstrated. In lieu of the 3-mile, 10 percent grade, brake fade test described above, the capability of the service brake system to deliver 70 percent full braking capacity to the brakes when applied at a rate of three applications per minute with transmission in high gear and engine at maximum governed speed shall be demonstrated.

4.7.5.7 Obstacle crossing ability. The truck with gross load shall be tested over an 14-inch vertical wall obstacle course. Any interference between vehicle components or between the vehicle and obstacle profile shall be noted and corrected. To further check for frame, suspension, and drive train flexibility, the truck at gross load shall be jacked up to allow placement of blocks under the wheels equal in height to the cross section diameter of the tires. The truck shall be in a static position and the blocks shall be placed under opposite wheels in alternately contrary directions.

4.7.5.8 Vehicle CI. To determine compliance to 3.6(i), the vehicle with gross load shall traverse terrain consisting of unflooded, fine-grained, dry-to-moist soils having CI rating in the critical layer no greater than 45* as specified in 6.5. Calculations will be acceptable to verify compliance.

4.7.5.9 Road and cross-country. The truck shall be subjected to two cycles of operation. One cycle with the water tank full and one cycle of operation with the water tank 50 percent full over the following roads and terrain at the speeds and distances specified:

Cycle	Average speed (mph)	Maximum speed (mph)	Distance (miles)
Hard surface highways	40	50	50
Gravel roads	25	35	30
Cross-country level	15	20	30
Cross-country hilly	15	20	30

4.8 First article environmental tests.

4.8.1 Environmental temperature test data. During the environmental operational tests, temperatures at least at the following points shall be recorded at intervals not greater than 15 minutes:

a. Ambient--two dispersed places.
b. Coolant inlet and outlet at booster heater.

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c. Coolant inlet for all major components and compartments being heated.
d. Air at center and bottom of all compartments being heated.
e. Air in cab at air inlet, operator's foot level, center, and ceiling.
f. Windshield at center.
g. Water at recirculating pump, and all nozzle shutoff valves.

4.8.2 Cold test. The truck shall be serviced with the proper low temperature greases, oils, and winter coolants; filled with fuel. The water system shall be drained and the drain time recorded. The truck shall be instrumented in accordance with 4.8.1 and the system inspected for undrained pockets of water. The truck shall be placed in a cold chamber using refrigerated air and the temperature within the chamber lowered and maintained at 0°±2°F for a period of 24 hours or until temperature stabilization occurs. The capability to start the engine at 0°F without use of the winterization system shall be demonstrated. The cold room temperature shall be lowered and maintained at -40°±2°F for a period of 24 hours or until temperature stabilization occurs. During the period, critical temperatures shall be recorded. At the completion of the cold soak period, the water tank shall be filled with at least 500 gallons of water and the vehicle's heating system activated (both engine, power plant heater, and the truck's heater system described in paragraphs 3.9.18.4 and 3.9.18.4.1). At the end of 1 hour operation of the vehicle's heating system, the capability of the truck to start within 15 seconds at -40°F shall be demonstrated. Also at this time, the vehicle firefighting system shall be checked by momentarily operating the water dispensing systems. Critical temperatures shall be recorded at this time. For a minimum period of 4 hours, at least two people shall occupy the cab to observe cab heating and window defrosting system operation, cab instrument operation, opening and closing cab windows and doors, and to record pertinent data and note any corrective action required for crew comfort during simulated standby operation for this period at -40°F. Also, during this test period a small amount of water shall be drained from each drain in the water system for evidence of any freezing conditions. At the end of the 4-hour period the water system shall again be momentarily operated. There shall be no evidence of freeze damage to any of the systems or components during the test.

4.8.3 Radio interference suppression test. To determine conformance to 3.18, the first article model shall be tested as specified in SAE J551 or the manufacturer shall furnish certified test data on an identically equipped vehicle that has been tested in accordance with SAE J551. Certified test data shall be accomplished by a description of all radio suppression devices used, adequate for visual determination that the first article model is identically equipped.

4.8.4 Noise level test. Noise level tests shall be conducted for compliance with 3.6.2.

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4.8.5 Rain tests. Rain tests specifically for demonstrating that cab construction is rainproof shall be conducted. The cab shall be tested to a water spray simulating a rainfall of 3 inches per hour for a period of 1 hour to determine leakage.

4.9 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample, and may be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

4.10 Production unit tests. Each production truck shall be subject to the following tests:

4.10.1 One hundred mile road test. The truck, with water tank 75 percent full, shall be operated for a distance of 50 miles during which 5 miles shall be in the 45 mph range; 20 miles at an average of 40 mph; 20 miles at a speed of 20-30 mph; and 5 miles in the 5-10 mph range. During the test, the following functions shall be performed:

a. Normal or average stops, 10 times.
b. Quick stops from a speed of at least 20 mph by instantaneous application of full service braking power, five times.
c. Sharp right turns and sharp left turns, five times each.
d. Operation through all forward and reverse gears including the transfer case if applicable.

4.10.2 Rejection and retest. When an item selected from production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the government representative the cause of failure and the action taken to preclude recurrence. After correction, all tests influenced by the failed components shall be repeated.

4.11 Packaging inspection. The vehicle shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Vehicle processing. Each truck shall be preserved and packaged in accordance with the contractors standard practice. When specified (see 6.2), the equipment shall be preserved and packaged in accordance with the mobile requirements of MIL-STD-281 with the level of preservation and packaging as specified (see 6.2).

6. NOTES

6.1 Intended use. The firefighting pumper tanker truck is intended to be used for assisting fire suppression at Naval Support Force, McMurdo Station, Antartica, South Pole.

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6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Specifications and standards (see 2.1.1).
c. When a first article inspection is required (see 3.2 and 4.2.1).
d. When accessory mounting brackets and holders are required (see 3.14).
e. When color of the finish coat and type of paint materials are specified (see 3.15).
f. When preservation and packaging shall be in accordance with MIL-STD-281; and the level of preservation and packaging required (see 5.1).

6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) and invokes the provisions of paragraph 52.227-7031 of the Federal Acquisition Regulations (FAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of FAR 52.227-7031 are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 First article. When a first article inspection is required (see 3.2 and 6.2), the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Definitions.

6.5.1 Cross-country terrain. Cross-country terrain consists of open fields, broken ground, loose sand or gravel road, and uneven terrain in general.

6.5.2 Vehicle CI. Personnel from the Waterways Experiment Station, U.S. Army Corps of Engineers, Vicksburg, Mississippi, have empirically developed a relationship between soil types and vehicle mobility with the aid of the cone penetrometer. The procedure for cone penetrometer measurements is reflected in the Department of the Army Technical Bulletin, TB ENG 37, dated 10 July 1959; and a means of predetermining vehicles mobility characteristics by computation is contained in U.S. Army Material Command Technical Memorandum No. 3-240 (18th supplement), date March 1968. In addition, vehicle CI requirements have been developed for a large number of representative U.S. Army vehicles. The following definitions pertain to this system:

a. Cone penetrometer. A device used in evaluating trafficability of soils. It consists of a 36-inch long metal shaft with a 30° right circular cone of 0.5 square inch base area mounted on one end of the shaft.

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b. Cone index. A dimensionless value read from the dial gage of the cone penetrometer. The CI is considered an index of shear resistance of the soil.

c. Vehicle CI. The minimum CI of a soil in the critical layer no greater than 45° which will permit a specific vehicle to make 50 passses without immobilization. It can be determined by first computing a mobility index from an empirical formula given in Appendix A of Technical Memorandum No. 3-240 and referring to a conversion table given in the same appendix.

d. Sheargraph. An instrument that measures surface traction properties of soils at normal pressures equal to the average tire contact pressures on a firm surface.

Preparing Activity:

Navy - YD

(Project 4210-N196)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-29237(YD)	TRUCK, FIREFIGHTING: PUMPER TANKER, 3,500 GALLON WATER CAPACITY DIESEL ENGINE DRIVE, 76,000 POUNDS GVW 6 BY 6

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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